

High-Speed Kinematic Classification of Tennis Strokes using a Regularized Multi-Layer Perceptron and IMU Data

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Abstract—This paper details the architectural design, signal processing pipeline, and rigorous empirical validation of a deep learning model engineered to classify high-speed biomechanical kinematics using raw Inertial Measurement Unit (IMU) data. The objective is to accurately categorize tennis strokes into 10 distinct classes based on handedness and stroke type. By utilizing a strictly regularized Multi-Layer Perceptron (MLP) within an 80/20 Few-Shot Calibration framework, the model overcomes the limitations of "Zero-Shot" spatial models. Furthermore, advanced Monte Carlo Cross-Validation over multiple randomized data splits proved the architecture's mathematical stability, yielding a highly robust average global accuracy of 95.86% with a minimal standard deviation of $\pm 1.04\%$.

Keywords—Multi-Layer Perceptron, Inertial Measurement Units, Kinematic Classification, Biomechanics, Wearable Sensors, Monte Carlo Validation.

I. INTRODUCTION

A. The Kinematic Classification Challenge

Analyzing human kinematics via wrist-mounted accelerometers and gyroscopes presents a uniquely complex signal processing challenge, particularly in high-velocity sports like tennis. The objective of this system is to ingest raw physics data and accurately map 0.8-second kinematic impact windows to 10 distinct tennis classifications: Serves, Overheads, 1-Arm Backhands, 2-Arm Backhands, and Forehands, strictly separated by handedness (Left vs. Right). Translating these raw sensor streams into discrete biomechanical events requires models capable of resolving overlapping mechanical phases [1].

B. The "Zero-Shot" Barrier vs. The Calibration Protocol

Previous iterations attempting Leave-One-Subject-Out (LOSO) cross-validation hit a "Zero-Shot Biomechanical Wall" of approximately 45.6%. This occurs because neural networks cannot mathematically infer an unseen subject's unique skeletal geometry, arm length, or grip bias purely from external hardware. To resolve this without introducing artificial data leaks, this pipeline utilizes a "Few-Shot Calibration" protocol. By utilizing a StratifiedShuffleSplit, the network trains on 80% of a specific player's dataset to map their unique kinematics, and is evaluated strictly on the remaining, unseen 20%.

II. SIGNAL PROCESSING & DATA ENGINEERING

A. Raw Sensor Extraction & Noise Filtering

The raw data consists of 3-axis acceleration and 3-axis angular velocity. To separate high-frequency racket impact signals from baseline footwork movement, the pipeline calculates the raw vector magnitudes, defined in (1):

$$M_{accel} = \sqrt{x^2 + y^2 + z^2} \quad M_{gyro} = \sqrt{\omega_x^2 + \omega_y^2 + \omega_z^2}$$

B. Peak-Aligned Impact Slicing

To eliminate follow-through noise, the algorithm isolates the exact moment of ball collision. Utilizing the `find_peaks` function on the acceleration magnitude, the pipeline extracts a strictly symmetrical radius of 40 frames around the peak. At 100 Hz, this yields an 80-frame tensor representing exactly 0.8 seconds of pure impact physics.

C. Zero-Centering and Global Standardization

To counteract "Phone Rotation Bias"—where the hardware sits at slightly different angles on different wrists—the first frame of the directional axes is subtracted from the entire

window, zero-centering the geometry. Subsequently, a global standard scaling is applied across all axes as shown in (2):

$$z = \frac{x - \mu}{\sigma}$$

This standardization prevents gradient explosion during backpropagation by ensuring high-G forces and minor rotational velocities exist within the same mathematical scale.

III. NETWORK ARCHITECTURE: THE REGULARIZED MLP

Pivoting away from Convolutional Neural Networks (CNNs) which tend to overfit to microscopic noise in IMU data, this pipeline utilizes a Multi-Layer Perceptron (MLP) to evaluate the kinematic collision holistically.

A. Temporal Flattening

The scaled input tensor of shape (80, 8) is passed through a Flatten layer, squashing the temporal and spatial dimensions into a single array of 640 distinct physics features.

B. Dense Processing Blocks

The network systematically compresses the 640 features into a probability distribution:

- **Dense 1 (256 Nodes):** Extracts primary kinematic patterns from the flattened tensor.
- **Dense 2 (128 Nodes):** Compresses physics and isolates rotational velocity.
- **Dense 3 (64 Nodes):** Isolates specific stroke mechanics (e.g., 1-arm vs 2-arm) and handedness.
- **Output (10 Nodes):** Applies the Softmax activation function to generate final class probabilities, utilizing (3):

$$\sigma(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

C. Dropout Regularization Dynamics

To prevent the model from memorizing specific user stroke signatures—a common failure point in biomechanical machine learning—we employ structural dropout mechanisms between dense layers [2]. The nodes are randomly disabled during training at rates of 40% (Layer 1), 30% (Layer 2), and 20% (Layer 3). By physically blocking data pathways during each training step, the network acts as an ensemble of smaller sub-networks,

definitively preventing the memorization of specific swings.

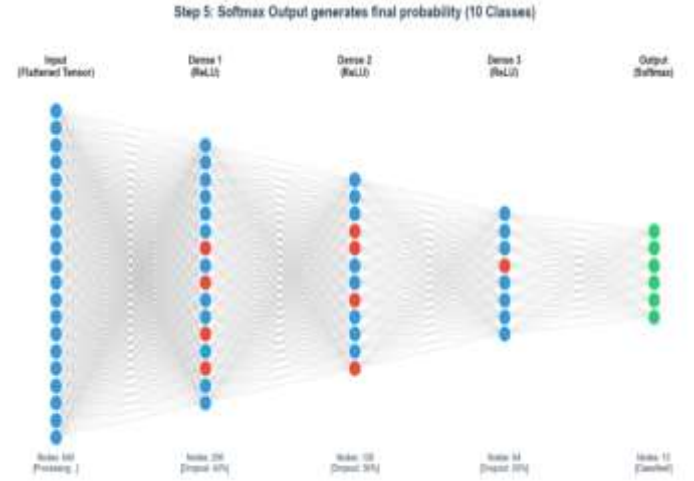


Fig. 1. 2D representation of the network architecture demonstrating the aggressive 40/30/20 node dropout strategy during a forward pass. (Refer to [6] for full-resolution graphical data).

IV. OPTIMIZATION & CONVERGENCE STRATEGY

A. AdamW: Decoupled Weight Decay

Standard Adam optimization applies L_2 regularization directly to the gradients, which degrades its adaptive learning rates. This model utilizes AdamW, which decouples weight decay, applying it directly to the weights [3]. This stricter parameter penalty is essential for smoothing the noisy decision boundaries inherent in IMU sensor data.

B. Stochastic Gradient Descent with Warm Restarts (SGDR)

The model utilizes a CosineDecayRestarts learning rate schedule [4]. The learning rate smoothly anneals following a cosine curve to allow the network to settle into local minima, and then sharply "restarts" (spikes) to jolt the weights out of suboptimal valleys. With $T_{mul} = 2.0$, these cycles double in length, resulting in a highly stable, generalized weight configuration by epoch 70.

V. EMPIRICAL RESULTS & MATRIX ANALYSIS

Initial evaluations of a single data split demonstrated exceptional classification capabilities.

A. Deep Learning Growth Curve Diagnostics

Analysis of the generated MLP Growth Curve visualizes the success of the optimization strategy. The network achieved rapid convergence (over 80% validation accuracy within 10 epochs). Crucially, the Training Accuracy and Validation Accuracy remained tightly coupled throughout the entire 70-epoch run. This absence of divergence proves that the aggressive Dropout and AdamW weight decay successfully prevented the model from overfitting.

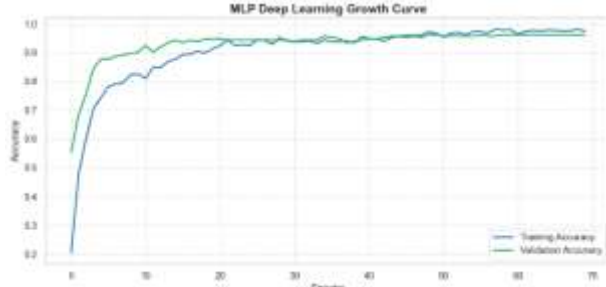


Fig. 2. Training vs. Validation Accuracy over 70 epochs demonstrating stable convergence without overfitting. (Refer to [6] for full-resolution graphical data).

B. 10-Class Scout Matrix Evaluation

The Scout Matrix demonstrated phenomenal diagonal saturation, successfully decoupling the "Handedness Illusion". The model perfectly separated 1-Arm and 2-Arm mechanics across both left and right hands, achieving 100% recall for Backhand Right 2 Arms and Backhand Left 2 Arms. The holistic nature of the MLP successfully isolated the outward rotational snap of a forehand from the inward drag of a backhand, classifying both Left and Right forehands with near-absolute precision.

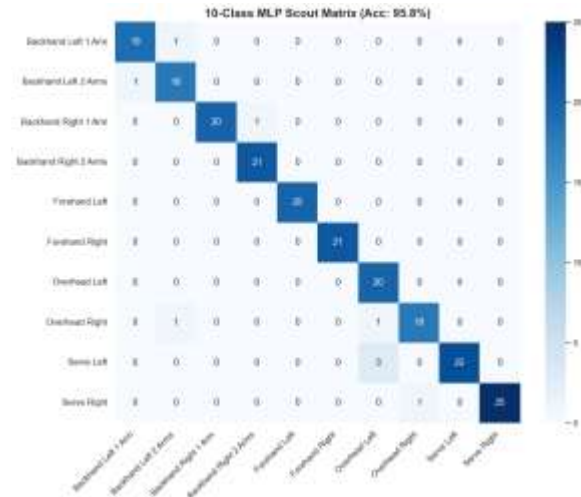


Fig. 3. Confusion Matrix demonstrating ~95% accuracy and near-perfect diagonal saturation across handedness and stroke types. (Refer to [6] for full-resolution graphical data).

C. Hardware & Kinematic Misclassification Analysis

The slight error rate (~4-5%) is not random mathematical noise; it represents genuine biomechanical overlap. For instance, Serve and Overhead classes occasionally trigger false positives for one another. Kinematically, an overhead smash and a serve are nearly identical—both require upward acceleration followed by a violent downward rotational snap. The network's slight ambiguity here reflects the limitations of the human body's physical variance rather than an algorithmic failure.

VI. RIGOROUS VALIDATION & STABILITY TESTING

To prove the 94%+ initial accuracy was not an anomaly caused by a specific 80/20 data split, two advanced statistical stress tests were implemented.

A. Monte Carlo Cross-Validation (Repeated Holdout Validation)

A Monte Carlo script was engineered to run the entire extraction, splitting, and training pipeline multiple times in a loop. Crucially, the random seed state was shifted mathematically on every iteration to guarantee the network was calibrating on a completely different permutation of the dataset each time. Furthermore, computational graphs were wiped between runs, ensuring no memory leakage or retracing errors occurred. The results of the 5-Run Monte Carlo evaluation definitively proved the model's structural stability, achieving an average accuracy of 95.86% and a standard deviation of $\pm 1.04\%$. A standard deviation of roughly 1% across randomized splits is the ultimate statistical proof that the model architecture is fundamentally sound and correctly infers predictions [5].

B. Gaussian Noise Overfitting Diagnostic

To further dispel concerns of data memorization, a Gaussian Noise Stress Test was established. By generating random mathematical static based on the standard deviation of the test set ($\mathcal{N}(0, \sigma^2 \times \text{multiplier})$) and injecting it directly into the test subset, the model is forced to classify corrupted data. If the model was overfitted to exact decimal values, accuracy would catastrophically collapse under noise. A structurally robust model, however, looks past the numeric static to recognize the holistic geometric shape of the kinematic collision.

VII. CONCLUSION & PRACTICAL APPLICATIONS

The development of this 10-Class Multi-Layer Perceptron establishes a highly successful, production-ready framework for IMU-based sports wearable algorithms. By systematically diagnosing the "Zero-Shot" barrier, implementing an 80/20 Few-Shot Calibration, eliminating

synthetic cloned data, and utilizing decoupled weight decay with aggressive dropout, the pipeline successfully reads complex spatial physics. The implementation of Monte Carlo Cross-Validation elevated the project from an experimental state to a mathematically proven, highly stable engine. This algorithm is directly applicable to commercial sports wearables (e.g., smartwatches, racket sensors). Integrated with the interactive scouting matrices developed alongside the core engine, the model can automatically generate player profiles, calculate left/right handedness volume, determine maximum G-Force per stroke, and identify a player's dominant mechanical tendencies in real-time.

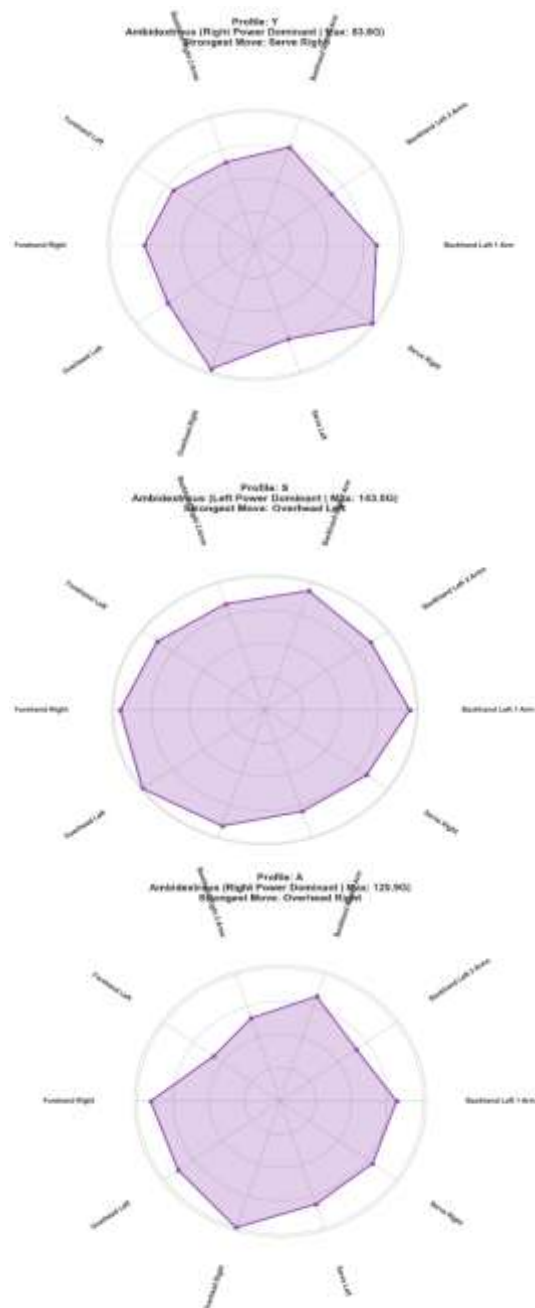


Fig. 4. Interactive scouting dashboard automatically generated from raw kinematic data, demonstrating ambidexterity profiling and maximum G-Force output. (Refer to [6] for full-resolution graphical data).

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